

One-Pager – IoT & AI Sensor Kits for the Kisii University Innovation Days 2026

Empowering Students with Practical IoT, Electronics & AI Skills

For the Innovation Days at Kisii University on April 7th, ChipGlobe Ltd. *i.f.* proudly donates **2×20 Carenuity Sensor IoT Boxes**. These kits enable 20 students to build their first IoT devices independently within minutes. Each kit comprises 2 boxes, which include the **C3-Mini microcontroller**, the ESP32-CAM microcontroller, a curated **set of modules like sensors**, an OLED display, relay, or buzzer, and the **Carenuity Triple-Adapter**, which seamlessly connects sensor, microcontroller, and actuator into many fully functional IoT units.

This donation is **currently unique worldwide** — no other university has received these boxes in this **special educational edition**. **Kisii University is the first institution globally** to receive and present these kits in this form, ahead of their official introduction at:

<https://www.carenuity.com/product/assortment-box/c3-mini-kisii-boxes>

The IoT kits will be officially presented and handed over by Timothy Mwala, the ChipGlobe representative for Kenya, who plays a key role in establishing the educational and innovation partnership between ChipGlobe Ltd. *i.f.* and Kisii University.

Students can instantly create devices such as barometers, hygrometers, thermometers, lux meters, IR radiometers, motion detectors or presence sensors — no prior experience required.

Fast, Browser-Based Development with the Solution Builder

All IoT kits are tightly integrated with the **Carenuity Solution Builder**, a fully browser-based development platform. It allows students to build and flash complete applications to the C3-Mini in **under three minutes**, without installing software, drivers, or development environments. Applications are selected, flashed via USB-C, and ready to run immediately. The platform provides transparent GitHub-hosted source code and a streamlined hardware-software integration process designed for both beginners and advanced developers.

This drastically lowers technical barriers and enables students to prototype, test and share ideas within minutes.

The Home Challenge – A Complete Learning Journey

The kits also give students access to the **Carenuity Home Challenge**, a structured education program covering the full spectrum of IoT skills — from soldering and simulation to PCB design, API integration, app development, 3D enclosure design, data logging and first machine-learning concepts. Each level builds on the previous one, offering a clear and motivating progression pathway for students of all backgrounds.

The program is already running in multiple countries and fits perfectly for developing emerging engineering talent in Kenya.

ChipGlobe Ltd. i.f. – Engineering Excellence for Emerging Innovators

ChipGlobe is a technology company specializing in **embedded systems, IoT, chip design, and prototyping**, with operational footprints across Europe and Asia. The company provides full development-flow expertise and offers embedded prototyping systems, microcontrollers and IoT hardware via its international web shop, with headquarters located in Munich, Germany.

ChipGlobe Ltd. i.f. aims to support Kisii University by:

- empowering hands-on engineering education in IoT & AI
- providing sustainable, scalable learning tools
- enabling students to create real, usable innovations
- building long-term collaboration between academia and industry
- positioning Kisii University at the forefront of global IoT education through this **first-of-its-kind gift**

Impact for Students

These Sensor IoT Boxes give students:

- **instant hands-on success** through plug-and-build hardware
- **practical, industry-relevant skills** in IoT, electronics, coding, APIs, data & AI
- **open tools, open source, and a global learning community**
- a **strong innovation foundation** for real-world problem-solving and future career paths

Together, the kits, the Solution Builder, and the Home Challenge form a complete ecosystem enabling students in Kisii to build meaningful IoT and AI solutions with global relevance.



Content of the Kisii-University-Boxes

The **two Kisii Boxes comprise** a 15-compartment box, named Box-1 and a 5-compartment Box, named Box-2)

Box-1: “Modular C3-Mini™ AI Product Dev. Box 1”

Box-2: “Modular C3-Mini™ AI Product Dev. Box 2”

Kisii STEM Education Boxes

Empowering Students Through Hands-On Engineering, IoT & Problem-Solving

The Kisii Boxes are curated, classroom-ready engineering toolkits designed to bring **modern STEM education, IoT prototyping, and real-world problem solving** directly to students. They support university labs, innovation hubs, and outreach programs in Kenya and beyond.

Two versions are available:

BOX-1: Kisii Engineering Box (15-Compartment Edition)

A **comprehensive engineering and IoT development kit** containing every core component needed for full-semester teaching, hands-on labs, and advanced student projects.

1. ESP32 Microcontroller Board (Wi-Fi + Bluetooth)

Description: A high-performance microcontroller with integrated Wi-Fi and Bluetooth, perfect for IoT, automation, and embedded systems.

Webshop: <https://www.chipglobe.shop/search?q=2404111-1>

Pedagogical Rationale: Enables computational thinking, embedded programming, and smart-device development, forming the core of modern engineering curricula. (Grover & Pea, 2013) [\[hackster.io\]](https://hackster.io)

2. Triple-Adapter (S-M-A)

Description: A modular adapter that simplifies connecting sensors, actuators, and displays to the ESP32.

Webshop: <https://www.chipglobe.shop/search?q=2411115-1>

Pedagogical Rationale: Reduces wiring errors and enables fast, structured prototyping aligned with best practices in cognitive scaffolding. [\[hackster.io\]](https://hackster.io)

3. DHT22 Temperature & Humidity Sensor

Description: Digital sensor for environmental monitoring projects.

Webshop: <https://www.chipglobe.shop/search?q=230199-1>

Pedagogical Rationale: Supports inquiry-based learning by enabling environmental data collection and climate-related experiments (Bybee, 2010). [\[hackster.io\]](https://hackster.io)

4. PIR Motion Sensor

Description: Infrared sensor detecting human movement, used in automation and security systems.

Webshop: <https://www.chipglobe.shop/search?q=19017-1>

Pedagogical Rationale: Lets students design smart-home systems, strengthening logic, abstraction, and real-world application (Papert, 1980). [\[hackster.io\]](https://hackster.io)

5. Light Sensor (LDR + 10kΩ Resistor)

Description: A photoresistor reacting to brightness changes.

Webshop: <https://www.chipglobe.shop/search?q=light+sensor+LDR>

Pedagogical Rationale: Connects physics concepts (light intensity) with programmable automation, supporting STEM integration (National Research Council, 2012). [\[hackster.io\]](https://hackster.io)

6. MQ-2 Gas Sensor

Description: Sensor for smoke, methane, LPG, and hydrogen detection.

Webshop: <https://www.chipglobe.shop/search?q=MQ-2>

Pedagogical Rationale: Enables safety-focused engineering projects and cross-disciplinary thinking (Kelley & Knowles, 2016). [\[hackster.io\]](https://hackster.io)

7. 2-Channel Relay Module

Description: Electromechanical relay board for switching higher-power devices with low-power microcontroller signals.

Webshop: <https://www.chipglobe.shop/search?q=relay+module>

Pedagogical Rationale: Introduces switching systems, control logic, and safe electrical engineering fundamentals (Kolb, 1984). [\[hackster.io\]](https://hackster.io)

8. 5V RGB Smart Bulb

Description: Programmable RGB LED module for UI design and feedback visualization.

Webshop: <https://www.chipglobe.shop/search?q=2301100-1>

Pedagogical Rationale: Reinforces design thinking and creative system feedback loops (Resnick, 2007). [\[hackster.io\]](https://hackster.io)

9. SG90 Micro Servo Motor

Description: Compact actuator enabling precise angular movement in robotics.
Webshop: <https://www.chipglobe.shop/search?q=2510200-1>
Pedagogical Rationale: Fundamental for robotics; teaches mechanical control, movement, and iterative testing (Eguchi, 2014). [\[hackster.io\]](https://hackster.io)

10. ESP32-CAM with OV2640 Camera Module

Description: Wi-Fi enabled camera for streaming, surveillance, and computer-vision tasks.
Webshop: <https://www.chipglobe.shop/search?q=230195-1>
Pedagogical Rationale: Supports digital literacy, ethics, computer vision, and data awareness (Livingstone & Helsper, 2007). [\[hackster.io\]](https://hackster.io)

11. Breadboard (830-Point)

Description: Solderless prototyping board for building electronic circuits.
Webshop: <https://www.chipglobe.shop/search?q=2501123-1>
Pedagogical Rationale: Promotes hands-on engineering and iterative experiment cycles (Kolb, 1984). [\[hackster.io\]](https://hackster.io)

12. Jumper Wires (65 pcs)

Description: Male-to-male connectors for breadboard prototyping.
Webshop: <https://www.chipglobe.shop/search?q=2510197-1>
Pedagogical Rationale: Teaches circuit design, debugging, and system interconnection. [\[hackster.io\]](https://hackster.io)

13. 0.96" OLED Display (Yellow-Blue)

Description: Small graphics display for sensor values, UI prototypes, and status messages.
Webshop: <https://www.chipglobe.shop/search?q=2411116-1>
Pedagogical Rationale: Builds understanding of data visualization and embedded UI systems. [\[hackster.io\]](https://hackster.io)

14. Buzzer

Description: Piezoelectric buzzer for audio signals and alerts.
Webshop: <https://www.chipglobe.shop/search?q=230198-1>
Pedagogical Rationale: Adds multimodal feedback and enhances engagement through sound-based output. [\[hackster.io\]](https://hackster.io)

15. SG90 Connector PCB

Description: Interface board to simplify clean servo connections.
Webshop: <https://www.chipglobe.shop/search?q=SG90+connector+pcb>
Pedagogical Rationale: Reinforces safe wiring, modularity, and electronics best practice. [\[hackster.io\]](https://hackster.io)

16. 15-Compartment Assortment Box

Description: Durable organizer for storing all components safely and neatly.

Webshop: <https://www.chipglobe.shop/search?q=2504139-1>

Pedagogical Rationale: Fosters organization, accessibility, and equitable resource distribution (UNESCO, 2017). [\[hackster.io\]](https://hackster.io)

BOX-2: Kisii Starter Box (5-Compartment Edition)

A compact, affordable introduction to IoT, electronics, and STEM learning. Contains a **carefully selected subset** of the Engineering Box — ideal for workshops, short courses, and mobile labs.

Items included:

1. ESP32 Microcontroller Board

Description: Wi-Fi enabled development board for IoT teaching.

Webshop: <https://www.chipglobe.shop/search?q=2404111-1>

Pedagogical Rationale: The fastest entry point into modern programming and IoT prototyping. [\[hackster.io\]](https://hackster.io)

2. Triple-Adapter (S-M-A)

Description: Adapter for connecting shields and sensors without wiring mistakes.

Webshop: <https://www.chipglobe.shop/search?q=2411115-1>

Pedagogical Rationale: Helps beginners succeed early by reducing wiring friction. [\[hackster.io\]](https://hackster.io)

3. DHT22 Temperature & Humidity Sensor

Description: Climate-sensing module for environmental projects.

Webshop: <https://www.chipglobe.shop/search?q=230199-1>

Pedagogical Rationale: Encourages data-driven science experiments. [\[hackster.io\]](https://hackster.io)

4. PIR Motion Sensor

Description: Detects movement for automation prototypes.

Webshop: <https://www.chipglobe.shop/search?q=19017-1>

Pedagogical Rationale: Connects real-world events with code logic. [\[hackster.io\]](https://hackster.io)

5. Light Sensor (LDR)

Description: Ambient light detector.
Webshop: <https://www.chipglobe.shop/search?q=light+sensor+LDR>
Pedagogical Rationale: Great for threshold logic exercises. [\[hackster.io\]](https://hackster.io)

6. MQ-2 Gas Sensor

Description: Smoke/gas detection component.
Webshop: <https://www.chipglobe.shop/search?q=MQ-2>
Pedagogical Rationale: Introduces safety engineering basics. [\[hackster.io\]](https://hackster.io)

7. RGB Smart Bulb

Description: Programmable LED light.
Webshop: <https://www.chipglobe.shop/search?q=2301100-1>
Pedagogical Rationale: Immediate visual feedback encourages experimentation. [\[hackster.io\]](https://hackster.io)

8. SG90 Servo Motor

Description: Mini servo for mechanical motion.
Webshop: <https://www.chipglobe.shop/search?q=2510200-1>
Pedagogical Rationale: Perfect for basic robotics tasks. [\[hackster.io\]](https://hackster.io)

9. ESP32-CAM Module

Description: Wi-Fi camera for image capture.
Webshop: <https://www.chipglobe.shop/search?q=230195-1>
Pedagogical Rationale: Inspires high-engagement computer-vision projects. [\[hackster.io\]](https://hackster.io)

10. Micro-USB Cable

Description: Power and programming cable.
Webshop: <https://www.chipglobe.shop/search?q=2501125-1>
Pedagogical Rationale: Ensures safe and reliable device operation. [\[hackster.io\]](https://hackster.io)

11. Breadboard

Description: Solderless prototyping platform.
Webshop: <https://www.chipglobe.shop/search?q=2501123-1>
Pedagogical Rationale: Ideal for fast classroom builds. [\[hackster.io\]](https://hackster.io)

12. Jumper Wires

Description: Wiring set for prototyping.
Webshop: <https://www.chipglobe.shop/search?q=2510197-1>
Pedagogical Rationale: Teaches practical debugging and circuit design. [\[hackster.io\]](https://hackster.io)

13. Buzzer

Description: Audio alert component.
Webshop: <https://www.chipglobe.shop/search?q=230198-1>
Pedagogical Rationale: Demonstrates sound-based output behavior. [\[hackster.io\]](https://hackster.io)

14. 0.96" OLED Display

Description: Mini graphical display.
Webshop: <https://www.chipglobe.shop/search?q=2411116-1>
Pedagogical Rationale: Helps visualize live sensor data. [\[hackster.io\]](https://hackster.io)
